

MIYAKO, Japan

WMO: 475850

Lat: 39.6471N

Lon: 141.9658E

Elev: 154

StdP: 14.61

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 20.4	(c) 22.8	(d) 4.5	(e) 7.0	(f) 28.4	(g) 6.8	(h) 7.9	(i) 28.8	(j) 18.3	(k) 35.3	(l) 15.0	(m) 34.8	(n) 5.5	(o) 240	(p) 0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 11.2	(c) 87.1	(d) 75.6	(e) 83.5	(f) 73.9	(g) 80.2	(h) 72.5	(i) 77.1	(j) 84.7	(k) 75.5	(l) 81.5	(m) 74.1	(n) 78.9	(o) 5.9	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 74.8	(b) 131.1	(c) 80.8	(d) 73.6	(e) 125.9	(f) 79.4	(g) 72.4	(h) 120.5	(i) 77.8	(j) 40.7	(k) 85.0	(l) 39.2	(m) 81.8	(n) 37.9	(o) 79.1	(p) 82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 14.6	(b) 12.3	(c) 11.1	(e) 15.9	(f) 92.8	(g) 2.9	(h) 2.1	(i) 13.8	(j) 94.3	(k) 12.1	(l) 95.5	(m) 10.5	(n) 96.7	(o) 8.5	(p) 98.2		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			14.4	80.0	2.7	1.5	12.4	81.1	10.8	82.0	9.3	82.8	7.3	84.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.0	33.0	33.5	39.0	48.2	56.7	62.1	69.6	72.7	67.0	57.0	46.6	37.6
	DBStd	14.77	4.39	5.00	5.86	6.81	6.50	5.53	6.19	5.47	5.48	5.36	5.88	5.07
	HDD50	1990	526	461	345	114	16	0	0	0	0	7	135	386
	HDD65	5289	991	881	805	505	266	120	24	4	38	253	552	850
	CDD50	2731	0	0	5	60	224	364	607	704	510	224	32	1
	CDD65	556	0	0	0	1	9	34	166	243	98	5	0	0
	CDH74	2841	0	0	0	15	75	162	845	1318	412	14	0	0
	CDH80	743	0	0	0	3	9	32	237	383	79	0	0	0
(14) Wind	WSAvg	5.1	5.3	5.4	5.7	5.9	5.4	4.6	4.0	4.3	5.0	5.5	5.4	5.2
(15) Precipitation	PrecAvg	51.4	2.6	2.7	3.3	3.4	3.8	4.4	5.6	6.5	8.2	5.0	3.5	2.0
	PrecMax	71.9	8.2	9.6	10.1	9.5	9.4	11.8	15.0	20.2	19.2	15.0	11.5	10.4
	PrecMin	32.5	0.1	0.1	0.5	0.2	0.4	1.2	0.6	0.8	1.1	0.3	0.1	0.0
	PrecStd	11.0	2.1	2.3	1.9	2.2	2.2	2.2	3.3	4.5	4.4	3.8	2.8	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.5	56.4	64.8	75.2	80.7	84.3	90.8	91.7	86.9	75.9	67.6	57.9
		MCWB	42.6	46.5	51.2	56.1	61.6	69.2	76.6	77.7	76.3	65.4	57.4	50.5
	2%	DB	46.1	49.7	58.7	69.0	75.5	78.5	86.1	87.9	81.9	71.6	63.5	53.8
		MCWB	39.3	41.3	47.6	53.8	60.6	66.9	75.3	76.8	72.9	63.2	55.0	47.1
	5%	DB	43.2	45.2	54.2	64.1	71.6	74.5	82.2	84.4	78.5	68.6	60.3	49.9
		MCWB	36.5	38.0	44.8	51.5	58.8	65.4	73.5	75.7	71.9	61.5	52.9	43.7
	10%	DB	40.7	42.1	49.8	59.8	67.8	70.9	78.5	80.8	75.6	66.2	57.4	46.9
		MCWB	34.5	35.8	41.9	49.9	57.4	63.9	72.3	74.3	70.4	60.1	51.2	40.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.3	47.5	52.9	60.8	66.1	72.0	78.3	79.5	77.4	69.0	59.5	53.0
		MCDB	48.5	54.7	61.0	68.3	74.7	78.1	86.3	87.9	84.1	72.7	64.2	56.2
	2%	WB	40.3	42.3	49.2	56.0	63.0	68.9	76.4	77.7	75.2	65.6	56.8	48.1
		MCDB	44.8	48.5	56.5	64.4	70.7	75.7	83.7	85.7	79.9	68.9	61.0	51.9
	5%	WB	37.7	39.1	45.9	53.6	60.9	66.7	74.6	76.2	73.4	63.3	54.4	44.5
		MCDB	41.9	43.9	53.0	61.3	68.6	72.6	80.5	83.0	77.6	66.8	58.7	49.7
	10%	WB	35.3	36.4	42.8	51.0	58.9	64.8	72.9	74.8	71.1	61.1	52.1	41.2
		MCDB	39.8	41.5	48.7	58.1	65.4	69.6	77.6	79.9	74.5	65.1	56.5	46.0
(35) Mean Daily Temperature Range	MDBR		13.8	14.5	15.7	16.9	15.6	12.4	10.9	11.2	12.5	15.7	16.9	14.5
	5% DB	MCDBR	16.6	19.4	23.2	24.2	23.6	19.7	17.8	17.2	16.0	17.7	20.8	18.5
		MCWBR	12.0	13.5	15.4	13.7	12.6	10.5	9.1	8.4	9.3	11.7	14.6	14.3
	5% WB	MCDBR	15.8	18.7	21.4	20.7	19.9	17.0	15.6	15.3	14.4	14.4	18.1	18.4
		MCWBR	12.5	14.1	15.7	13.2	12.0	10.1	8.8	8.0	8.9	10.8	14.2	15.2
(40) Clear-Sky Solar Irradiance	taub		0.371	0.419	0.500	0.549	0.567	0.546	0.549	0.526	0.468	0.435	0.405	0.378
	taud		2.131	1.986	1.836	1.791	1.811	1.938	2.005	2.069	2.189	2.162	2.144	2.142
	Ebn at Noon		241	244	236	235	234	239	237	239	245	238	227	226
	Edh at Noon		35	47	60	67	67	59	55	50	42	39	35	32
(44) All-Sky Solar Radiation	RadAvg		634	900	1201	1485	1592	1563	1398	1387	1165	941	672	534
	RadStd		41	53	67	128	152	107	176	174	103	89	44	43

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page